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SUPERSEDING
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MILITARY SPECIFICATION

GRINDING MACHINES, JIG, VERTICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers floor mounted, vertical spindle, precision, jig grinding machines with equipment and accessories specified herein.

1.2 Classification. This specification covers machines of the following types, styles, and sizes. The type, style, and size machine required for specific procurements shall be as specified (see 6.2.1).

Type X - Manual Positioning

** Size	Table Travel (inches)
1	18 X 11

Type II - Power Positioning

Style A - Single Column

** Size	Table Travel (inches)
1	8 x 8
2	16 x 10
3	18 x 11

Style B - Double Column

** Size	Table Travel (inches)
1	25 x 18
2	32 x 24
3	36 x 24
4	48 x 24
5	48 x 24 (Extended table size)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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** When another size is required, machine size and corresponding characteristics shall be as specified (see 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/NEMA MG 1	- Motors and Generators.
ANSI/NFPA 79	- Metalworking Machine Tools and Plastics Machinery, Electrical Standards for.
ANSI B7.1	- Abrasive Wheels, Use, Care, and Protection of, Safety Codes for the.
ANSI B11.9	- Safety Requirements for the Construction, Care and Use of Grinding Machines.
ANSI B74.13	- Marking for Identifying Grinding Wheels and Other Bonded Abrasives.
ANSI B74.2	- Shapes and Sizes of Grinding Wheels, and Shapes, Sizes, and Identification of Mounted Wheels, Specification for.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951	- Commercial Packaging, Standard Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION

AGMA 360.02	- Manual For Machine Tool Gearing.
AGMA 390.03	- Gear Classification, Materials and Measuring Methods For Unassembled Gears.

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(Application for copies should be addressed to the American Gear Manufacturers Association, Standards Department, Suite 1000, 1901 North Fort Myer Drive, Arlington, VA 22209-1695).

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

ISO R1122 - Gears, Glossary of, Geometrical Definitions.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Design. The machine shall be new and one of the manufacturer's current models capable of locating and grinding cylindrical and conical shaped holes and combinations of radii, tangents, angles, and flats in hardened steel and other metals and materials in accordance with the requirements herein. Type I machines require manual table positioning. Type II machines position automatically by power to table coordinates determined by the operator. Style A machines are of single column construction; style B machines are of double column construction. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.1.1 Measuring systems. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.1.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indica-

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tors, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments, and dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.1.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use or performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.1.3 Energy efficiency. The machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.1.4 Operating controls. All operating controls necessary for manual or power operation of the machine shall be mounted on either a control console or pendant stand clearly identified to facilitate its use. Voltage and current meters, indicator lights, and all other operational control aids shall be arranged and mounted convenient to and in full view of the operator. An emergency stop switch shall be furnished on the control.

3.1.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with ANSI B11.9 and with all of the requirements of OSHA 2206 that are applicable to the machine itself. In the event of a conflict between the requirements of the ANSI and OSHA standards, the OSHA standards shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.1.6 Mercury restriction. The machine shall not contain mercury or mercury components nor be exposed to free mercury during manufacture.

3.1,7 Asbestos restriction. Asbestos and materials containing asbestos shall be not be on or in the machine.

3.1.8 Environmental protection. The machine shall be designed and constructed such that, under the operating, service, transportation and storage conditions described herein, the machine shall not emit materials hazardous

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to the ecological system as prescribed by Federal, state or local statues in effect at the point of the installation.

3.1.9 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification to the part or the machine.

3.2 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.2.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on the machine. Such processes may be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.2.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part .

3.2.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.2.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins , sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.2.6 Threads. Except for the leadscrew, when applicable, all threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.2.7 Dials. All dials that indicate stock removal or tool movement shall be g-cd in increments not larger than 0.001 inch. Dial diameters shall be such that graduations may be read from the normal operating posi-

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tion. Dials shall be permanently and legibly engraved or etched on a non-glare background.

3.2.8 Gears. All gears used in the machine or its components shall be properly selected for the intended application and fabricated to either English (USA) standards per AGMA 390.03 and AGMA 360.02 or Metric (SI) standards per the International Organization for Standardization ISO R1122 and ISO 54.

3.2.9 Machine ways. The machine ways shall be adequately protected whenever possible to exclude grinding dust, coolant, and other hazards that may cause damage or result in inaccurate functioning of any members. The machine ways shall be hardened and ground and suitable for the high precision machine covered by this specification.

3.2.10 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.3 Components. The machine shall consist basically of the following components . Any additional components necessary for the operation of the machine shall be furnished in accordance with manufacturer's standard practice.

3.3.1 Base. The base shall be a casting of iron, iron alloy or a fabricated weldment, ribbed and braced to minimize distortion, vibration, and deflection. The base shall support the column(s), bridge, spindle housing assembly, worktable, saddle, and cross carriage, as applicable. The coolant, hydraulic and electrical systems shall be separate external units. When applicable, an access door(s) or removable plate(s) shall be provided to service units housed within. Guideways shall be provided to accommodate either a sliding saddle or worktable, as applicable. Means shall be provided to level the machine.

3.3.2 Column assembly. Each column shall be a one-piece casting of box-like construction, internally ribbed and braced to minimize distortion and deflection under normal loads. The column on type I and type II, style A machines shall accommodate the main spindle housing assembly and its associated components. Guideways shall serve to support and guide the spindle housing assembly through its full range of vertical adjustment. Means shall be provided to protect column ways from dirt+ Means shall be provided for vertical adjustment of the spindle housing assembly with a device to securely lock the spindle assembly in any location within its vertical range. A graduated scale or some other visible means shall be provided to return the spindle housing assembly to the same location after assembly movement. The machine shall have either stops or limit switches located at the extremes of spindle housing assembly motion to prevent excess travel of the spindle assembly. Type II, style B machines shall meet the same basic requirements as the style A machines except that they shall have a double column, bridge

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and sliding cross carriage configuration accommodating the main spindle assembly.

3.3.3 Spindle housing assembly. The spindle housing assembly shall include, but not be limited to, the main drive spindle and the grinding spindle or head with corresponding feed and speed mechanisms, vertical slide handwheel, spindle housing clamp, adjustable depth stop, heating elements, and means for indicating housing temperature. The assembly shall be made of a nickel iron alloy with a low coefficient of thermal expansion. Means shall be provided in the spindle housing assembly to simulate bearing running heat when the spindle is not running but the machine is operative to minimize the effects of expansion and contraction due to temperature changes common with variations in spindle operation. The assembly shall be mounted on the column or cross carriage guideways, as applicable, and shall traverse vertically (W axis). The assembly shall be balanced and counterbalanced. A non-influencing clamping mechanism shall be provided to lock the assembly to the column at any position within its vertical travel. The main spindle shall be of steel alloy and shall rotate in preloaded ball or roller bearings. The motor that drives the grinding spindles or heads shall be either an electric or pneumatic type. Provisions shall be made to adjust the spindle speed in either continuous adjustment or in steps of not greater than 10 percent. The grinding head shall be designed to accommodate grinding wheels mounted on 3/8 and, when specified (see 6.2.1), 1/4, and 1/8 inch shanks using adapters, if required. Means shall be provided to indicate vertical spindle housing assembly location.

3.3.3.1 Taper grinding. A taper grinding attachment shall be provided to allow grinding of convergent and divergent tapers. A taper–indicating dial or plate shall be provided to set the angle of inclination throughout its entire range. The indication dial shall have a maximum graduation of 30 minutes.

3.3.3.2 Radial grinding wheel off-set. Provision shall be made for horizontal adjustment or off-setting (U axis) of the grinding wheel from the main spindle’s fixed axis of rotation. A rough adjustment shall be provided to allow the grinding wheel to be offset to the approximate required dimension while the machine is inoperative. A fine adjustment shall be provided to adjust the grinding wheel to the exact dimension while the machine is in operation. The fine adjusting off-set mechanism shall have a dial to indicate offset with a maximum graduation of 0.0001 inches.

3.3.3.3 Axial reciprocation. Power reciprocation of the grinding spindle shall be provided. Manual operation, as required by the operator for set-up, will be through the use of a handwheel conveniently located near the operator. Power vertical reciprocation of the grinding spindle shall be provided with a pneumatic compensating device or counterbalance. An adjustable positive stop shall be provided to set lower travel limit to prevent overtravel. A dial or scale shall indicate spindle position, graduated in increments not larger than 0.001 inches.

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3.3.3.4 Contour grinding. Provision shall be made to grind cylindrical sections between predetermined angular limits. It shall be possible to swing the spindle back and forth through a controlled arc. A direct reading indicator or scale shall be provided. Adjustable stops shall be provided so that an arc of any desired size within the range of the machine can be established without danger of overtravel. A mechanism to swing the spindle manually through the arc defined by the stops shall be provided.

3.3.3.5 Planetary movement of grinding wheel (C axis). Provision shall be made for planetary movement (C axis) of the grinding wheel. The grinding wheel shall not only rotate about its own axis but shall have circular or planetary movement so that the eccentrically offset wheel follows around the contour or hole to be ground. Planetary offsetting shall be manually controlled, utilizing a handwheel.

3.3.4 Worktable. The worktable shall be a casting of iron or iron alloy, designed and proportioned to withstand any loads that may be encountered during normal operation. Unless otherwise specified (see 6.2.1), the number, size and configuration of T-slots in the table shall be manufacturer's standard. The surfaces of the worktable shall be hand or machine finished for a work mounting surface. When flood coolant is supplied (see 3.3.10), the top of the table shall have coolant troughs or similar means for draining spent coolant into its reservoir. The table shall be provided with a means of securing workplaces in place to permit precision grinding without drifting. The worktable shall be mounted on a traversing saddle on style A machines and shall travel in the X-axis direction. On style B machines , the worktable shall be mounted on the base and shall traverse in the X-axis direction. The worktable shall be provided with protective shields or covers for leadscrews, ballscrews, motors, and guideways, as necessary, to guard against the entry of foreign material. Means shall be provided at each end of travel to prevent table overtravel. Type I machine functions shall be controlled manually. Worktable functions on type II machines shall be controlled by power.

3.3.5 Saddle. Type I and type II, style A machines shall be provided with a saddle. The saddle shall be a casting of iron or iron alloy designed and proportioned to withstand all loads which may be imposed during normal operation of the machine. The saddle shall be mounted on the base guideways and shall traverse in the Y-axis direction. Longitudinal guideways shall be provided on the saddle top to accept a sliding worktable. The saddle shall operate on a leadscrew or precision ballscrew with a recirculating ball nut. Protective shields shall be provided for leadscrews, ballscrews, motors, and guideways, as applicable, to guard against the entry of foreign matter. Means shall be provided to prevent movement of the saddle after positioning. Stops shall be provided at each end to prevent overtravel. Saddle functions on type I machines shall be controlled manually. Saddle functions on type II, style A machines shall be controlled by power.

3.3.6 Bridge. Type II, style B machines shall be constructed with a cast iron, bridge-like member. It shall be internally ribbed and braced to mini-

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mize distortion and deflection experienced under normal operating loads. The bridge shall be fixed, supported on two vertical box-type columns located on either side of the base casting, simulating a bridge over the base and worktable. Horizontal guideways shall be provided on the bridge to accept a moveable cross carriage assembly and to allow the cross carriage to move in the Y-axis direction. On machines with a cross rail, vertical guideways shall be provided in the columns supporting the cross rail to allow the cross rail to traverse in the vertical (W-axis) direction. The cross rail shall be equipped with fully automatic, distortion-free clamping. Horizontal guideways shall be provided the cross rail to accept a cross carriage and to allow the cross carriage to move in the Y-axis direction. The cross rail shall be pneumatically balanced to assure parallel movement of the spindle headstock in relation to the worktable. Covers or protective shields shall be provided to guard against the entry of foreign material to leadscrews, ballscrews, motors, and guideways. Stops shall be provided to prevent overtravel.

3.3.7 Cross carriage. Type II, style B machines shall be provided with a cross carriage. The cross carriage shall be made of iron or iron alloy, precision fitted to the bridge or cross rail, traversing in the Y-axis direction. The cross carriage on machines without a cross rail shall be provided with vertical guideways to accommodate the spindle housing assembly and its associated components and to allow the assembly to traverse vertically (W-axis). On machines with a cross rail, the cross carriage and spindle housing assembly shall travel in the Y-axis direction on horizontal guideways on the bridge and vertically (W–axis) on vertical guideways in the columns. Aprons or covers shall be provided to guard the guideways from contamination. Means shall be provided to secure cross carriage location after positioning. The cross carriage shall be provided with power drives.

3.3.8 Grinding wheels. Unless otherwise specified (see 6.2.1), grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1.

3.3.9 Lubrication. A lubrication system shall be provided which will insure adequate lubrication for all moving parts. The system shall deliver lubrication to all leadscrews, ballscrews, mating gears, sliding parts, and bearings except the sealed-for-life type. Each lubricant reservoir shall have a means for determining and maintaining fluid level. The lubricant reservoir shall have sufficient capability to operate for a minimum of 80 hours of feed time without being serviced. The lubrication system shall have a device for indicating oil level and shall be readily visible to the operator. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. A recommended preventive maintenance schedule shall be provided.

3.3.10 Coolant system. Unless otherwise specified (see 6.2.1), the machine shall be supplied with a coolant system. The system shall have adequate capacity for all types of grinding operations for which the machine

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is designed and shall have means for regulating the flow of flood coolant or for adjusting mist coolant supply, as applicable. All components necessary to make the system completely functional shall be furnished including reservoirs, splash guards, filters, and drain troughs, as applicable.

3.3.11 Hydraulic system. The hydraulic system, when supplied, shall be complete with pumps, valves, piping, cylinders, reservoir, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. The system shall include filters to insure delivery of clean fluid. The reservoir shall have a sight gage to indicate fluid level. The fluid temperature shall be maintained within the proper operating range during all normal operations of the machine.

3.3.12 Pneumatic system. Unless otherwise specified (see 6.2.1), a pneumatic system shall be provided. Shop air shall be provided having a line pressure of 90 to 120 psi. The machine shall have an air line filter, water trap, and oil mist lubricator, as necessary. The system shall include all devices such as pressure regulators, pressure reducers, solenoid valves, and check valves that are necessary to meet the requirements specified herein. Air pressure gauges shall be provided to indicate input air pressure in all lines that are downstream from a pressure control device.

3.3.13 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), the machine shall be operable on 440 volts (with an allowable line fluctuation of +10 percent), three phase, 60 Hz source and shall be initially wired for operation on 440 volts. It shall be possible to operate all electric equipment on the machine after connection to a single power source.

3. 3.13.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet he requirements of dripproof enclosure in accordance with ANSI/NEMA MG-1. Pneumatic motors shall be of the vane or impulse reaction turbine air motor type controlled by the regulation of air pressure (see 3.3.12).

3.3. 13.2 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 115 volts.

3.3.14 Drive mechanisms. Means shall be provided to position each moving member in all axes of motion. All leadscrews or ballscrews shall be hardened and precision ground, and be of a size, class and precision suitable for the application. Leadscrews or ballscrews and nuts shall not exert excessive loads or cause premature deterioration of the feed drive components. Means shall be provided to insure proper lubrication and minimal thermal contraction and expansion of the screw. The slides and screws shall be protected from foreign elements by covers. This system, through operator

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control of each axis, shall enable the drives to have the ability to rapid traverse and decelerate to the actual desired positions.

3.3.14.1 Manual positioning. Type I machines shall position along each coordinate axis by manual means, utilizing handwheels and precision ballscrews or leadscrews. Movement shall be initiated and controlled by the operator.

3.3.14.2 Power positioning. Type II machines shall position along each coordinate axis by power. For positioning, the system shall require that the operator, by single push button, lever, or selector switch, located the spindle by power to a chosen position on the workpiece. The system shall be permanently built into the machine in a manner to prevent entrance of dust, swarf, or other foreign matter onto surfaces which might affect the accuracy of positioning. On style A machines, longitudinal table travel (X axis), cross axis saddle movement (Y axis), and spindle housing assembly vertical travel (W axis) shall be controlled by power. On style B machines, cross carriage travel (Y axis), table travel (X axis), and cross rail vertical travel (on moveable cross rail machines) or spindle housing assembly vertical travel (W axis) shall be controlled by power. Axial reciprocation of the grinding spinding (Z axis) and radial infeed of the grinding wheel (U axis) shall also be controlled by power. Machine set-up for taper grinding, contour grinding, and planetary movement of the grinding wheel (C axis) and operation of a rotary table (A axis), as applicable, shall be controlled manually by the operator. Rectilinear movement of the X, Y and W axes shall be accomplished by solid state servo drives, electric motors, position and velocity feedback devices, and precision leadscrews or precision, preloaded, antifriction recirculating ballscrew and nut assemblies forming a closed loop servo system.

3.3.15 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of O to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically revert to zero and continue to total time.

3.3,16 Digital readout system. When specified (see 6.2.1), a digital readout system shall be provided. The digital readout system shall consist of a position feedback device and electronic display for each axis specified (see 6.2.1) and all related components, mounting hardware, and electrical equipment necessary to assure complete operation of the system. The system shall be capable of providing a readily visible display over the entire working range of each machine axis on which installed. The system resolution shall be 0.00005 inch. The system accuracy shall be 0.0002 inch per foot and repeatability shall be 0.0001 inch. The readout system shall be capable of indicating axis location in both the US and SI systems of measurement.

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3.3.17 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device during operation of the machine. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis.

3.3.18 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with the machine:

a. One set of special wrenches, operating and repair tools that are normally furnished by the manufacturer.

b. Protective shield with an adjustable mounting arm.

c. One worklight, adjustable to illuminate work area, complete with protective shield and on-off switch.

d. Work holding and mounting devices as follows:

(1) One set of rests for height build-up from not more than 3/8-inch to at least 9 inches.

(2) One set of clamps having adjusting screw at heel end; adjustment shall be variable through at least one inch.

(3) One set of extension bolts, complete with washers and nuts, for make-up to any height within the range of the machine.

e. The required number of indicator sets, complete with clamps, to indicate holes from 3/16-inch to 7-1/2-inch diameter.

f. Grinding wheels: shank size, number of each wheel size, type, dimensional characteristics, and composition shall be as specified (see 6.2.1).

3.3.19 Optional equipment. The following equipment and accessories are commercially available and are considered optional. When required, the quantity of each item and its characteristics shall be as specified (see 6.2.1):

a. Storage cabinet.

b. Microscope.

c. Collet type adapters for grinding heads.

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d. Collet sets for grinding arbors.

e. High power 9,000 RPM grinding head with removable arbors of different lengths.

f. Dust exhaust system.

g. Standard 40,000 RPM grinding head.

h. Medium speed 60,000 RPM grinding head.

i. High speed pneumatic 100,000 RPM grinding head.

j. Super high speed pneumatic 160,000 RPM grinding head.

k. Precision rotary table.

l. Precision spin table.

m. Wheel dresser, as specified (see 6.2.1).

n. Parallel set-up blocks,

o. Horizontal spindle slot grinder.

p. Extension plates.

q. Angle, wedge, or aligning plates.

r. Other optional equipment as specified (see 6.2,1).

3.4 Sizes and capacities. Unless otherwise specified (see 6.2.1), the sizes and capacities of the machine shall meet the requirements of table I as applicable for the machine size specified in the contract or order.

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TABLE I. Sizes and capacities.

Characteristics	Type I	Type II		
		Style A		
	Size 1	Size 1	Size 2	Size 3
CAPACITY:				
Table:				
Longitudinal (X) travel (inches)	18	8	16	18
Cross (Y) travel (inches)	11	8	10	11
Work surface (inches)	24 x 11	15 x 11	23 x 15	24 x 11
Table top to wheel collet, adjustable (inches)	2 to 18	2 to 8-5/8	0 to 19-3/4	2 to 18
Permissible table load (lbs)	300	220	660	300
Bridge:				
Distance between columns (uprights)	—	—	—	—
Spindle:				
Spindle to column guideways (inches)	10	10-1/4	13-3/8	10
Vertical spindle travel (inches)	3-5/8	1-1/4	4-3/4	3-5/8
Vertical head assembly travel on column (W axis) (inches)	12-5/8	10	18	12-5/8
Grinding capacity (inches)				
-without extension plates	3-1/2	up to	9	3-1/2
-with extension plates	13-1/2	5/8	14	13-1/2
Radial offset (U) of grinding wheel:				
Rough adjustment (inches)	← ±.75 →			
Fine adjustment (inches)	← .25 →			
Angular adjustment from spindle centerline (for taper grinding) each side of centerline	0° to 1-1/2°	N/A	0° to 8°	0° to 1-1/2°
FEEDS AND SPEEDS:				
Number of main spindle speeds	← Infinite →			
Main spindle speed range (minimum to maximum) (RPM)	25 - 225	370	5-250	25 - 225
Grinding spindle speed range using full range of heads (RPM)	6000 to 160,000	100,000 and 160,000	← 6,000 to 160,000 →	
Vertical reciprocating				
Frequency of stroke (number)	← Infinite →			
Range (IPM)	4 to 240	1/4 to 37	← 4 to 240 →	
Range of stroke length (minimum to maximum) (inches)	1/10 to 3-5/8	1/10 to 1-1/4	1/10 to 4-3/4	1/10 to 3-5/8
Axis (X and Y) feed rate (IPM)	None	0 to 28	59	120
Machine net weight (lbs) (without electrical cabinet)	3980	1760	6160	3980

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TABLE I. Sizes and capacities. Continued.

Characteristics	Type II Style B				
	Size 1	Size 2	Size 3	Size 4	Size 5
CAPACITY:					
Table:					
Longitudinal (X) travel (inches)	25	32	36	48	48
Cross (Y) travel (inches)	18	24	24	24	24
Work surface (inches)	30 x 24	36 x 22	40 x 33	48 x 24	52 x 33-7/8
Table top to wheel collet, adjustable (inches)	0 to 23-1/2	6 to 24-1/2	0 to 29-1/2	6 to 24-1/2	0 to 29-1/2
Permissible table load (lbs)	1100	800	1764	2000	1764
Bridge:					
Distance between columns (uprights)	29-1/2	27	37-3/8	32	37-3/8
Spindle:					
Spindle to column guideways (inches)	—	—	—	—	—
Vertical spindle travel (inches)	5-1/2	5	7-3/32	5	7-3/32
Vertical head assembly travel on column (W axis) (inches)	19-1/2	13	25	13	25
Grinding capacity (inches)					
-without extension plates	9	5-1/4	9	5-1/4	9
-with extension plates	14	13-1/2	14	13-1/2	14
Radial offset (U) of grinding wheel:					
Rough adjustment (inches)	← +.75 →				
Fine adjustment (inches)	← −.25 →				
Angular adjustment from spindle centerline (for taper grinding) each side of centerline	0° to 8°	0° to 1-1/2°	0° to 8°	0° to 1-1/2°	0° to 8°
FEEDS AND SPEEDS:					
Number of main spindle speeds	← Infinite →				
Main spindle speed range (minimum to maximum) (RPM)	5 - 250	20 - 180	5-250	20 - 180	5 - 250
Grinding spindle speed range using full range of heads (RPM)	← 6,000 to 160,000 →				
Vertical reciprocating					
Frequency of stroke (number)	← Infinite →				
Range (IPM)	← 4 to 240 →				
Range of stroke length (minimum to maximum) (inches)	1/10 to 5-1/2	1/10 to 5-1/8	1/10 to 7-3/32	1/10 to 5-1/8	1/10 to 7-3/32
Axis (X and Y) feed rate (IPM)	59	16	98	16	118
Machine net weight (lbs) (without electrical cabinet)	8800	10,000	13,200	12,000	15,400

NOTE: Machines sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size as shown in above table. When the largest size and capacity machine of the machine shown in the above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by-more-than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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3.5 Performance. Each machine shall be capable of performing grinding operations to tolerances within +0.0001 inch on size, 0.00005 inch on roundness , and locational accuracy between holes of +0.00009 inch for type I machines and +0.00032 inch for type II machines.

3.6 Alignment tolerances. Each machine shall meet the alignment tests in 4.6.3.

3.7 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.8 Marking of instruction plates, indicating plates and nameplates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.8.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to the machine. If a chart is furnished, it shall be placed in a transparent plastic folder or permanently sealed between clear plastic sheets with suitable means for mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.8.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the Information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer’s permanent records. The captions listed may be abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer’s name
Manufacturer’s model designation
Manufacturer’s serial number
Power input(volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.9 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer’s commercial equipment of the type specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).

b. Acceptance test (see 4.4).

4.3 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.5 and the inspection in 4.7. Failure of the item to pass the examination or tests shall be cause for rejection.

4.4 Acceptance test. An acceptance test shall be performed on the grinder to insure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall be performed only after the grinder is installed at its final location and shall consist of all tests in 4.6. Failure of the grinder to pass one or more tests or any part of a test will be cause for rejection.

4.5 Examination. The grinder and equipment shall be examined for design, construction, material, components, electrical equipment and workmanship to determine compliance with the requirements in 3.1 through 3.4, 3.8 and 3.9.

4.6 Tests. The machine shall be subjected to the following tests. All instruments, tapes, materials, and tools required to perform and evaluate these tests shall be furnished by the supplier.

4.6.1 Operational test. The machine shall be operated at no load for not less than one hour. The machine shall be operated in accordance with the manufacturer's standard operating test procedure for warm-up and run-off checks. During the warm-up period, proper operation of all controls, motors, adjustment mechanisms, and accessories shall be verified.

4.6.2 Performance tests. Unless otherwise specified (see 6.2.1), the following test shall be performed to demonstrate the machine's capability to accurately position, grind and relocate. Prebore three 1.005 inch diameter

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through holes in an equilateral triangular pattern in a metal plate not less than one inch thick. (If the given test dimensions exceed a machine's capacity, prebore three 0.5005 inch diameter through holes in an equilateral triangular pattern in a metal plate not less than one-half inch thick.) The plate shall be of material chosen by the supplier. One or both faces of the plate may be surface ground prior to the test. The dimensions between bored holes shall be within one inch of the limits of cross-axis travel of the sliding members. Finish grind the holes to a diameter of 1.0170 inch. Maximum deviation on size shall be not more than 0.0001 inch. Remove the workpiece from the machine and relocated it on the worktable rotated 90 degrees. Using a locating dial indicator and the measuring system of the machine, relocate the ground holes. The machine shall accurately relocate the second and third holes to within plus or minus 0.00009 inch for type I machines and within plus or minus 0.00032 inch for type II machines. The machine's positioning shall be determined by comparison to a linear measuring instrument with calibrated accuracy directly traceable to the National Bureau of Standards.

4.6.3 Alignment and accuracy tests. Each machine and its components shall be tested in the assembled machine for alignment and accuracies. Unless otherwise specified (see 6.2.1), the alignment and accuracies shall be in accordance with table II. The instruments and methods used in performing the tests shall be at the option of the supplier provided they accurately determine the tolerances of the machine. Tests shall be performed in an ambient temperature which differs from 70° F by not more than 5°.

4.6.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.2.10.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall conform to the requirements of ASTM-D-3951. When specified (see 6 .2.1), level A packaging, level A or level B packing, and special marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The jig grinding machines covered by this specification are intended for use in any manufacturing shop where precision grinding of jigs, dies, and other similar precise work is performed. They will be used to locate and grind cylindrical holes, conical holes with tapers in either direction, and contours including combinations of radii, tangents, angles, and flats.

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TABLE II. Alignment and accuracy requirements.

Characteristics	Maximum allowable deviation
Squareness between X and Y axes	0.0001" per 12"
Trueness of table travel, rise and fall within	0.0001" per 12"; 0.0002" total
Trueness of saddle or cross carriage travel, as applicable, rise and fall within	0.0001" per 12"; 0.0002" total
Table flatness, checked with indicator in the spindle, measured at 5" by 5" grid points over entire table	0.0002" TIR
Table sweep, at 4" radius from spindle axis	0.0003" TIR
Head and spindle movement square with table; 12" high test cylinder on table, check both X and Y planes	Spindle 0.0002" per 12" Head 0.0002" per 12"
Grinding test - 2" to 3" diameter hole in 6" square meehanite block 1-1/2" thick	Round within 0.000050" TIR on diameter
Repeatability (at one position)	10 cycles, ±0.00004"
Positioning accuracy	0.0001" per 12" 0.0002" overall

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, style, and size required (see 1.2).

c. Units of measurement system to be used in the design and construction of the machine (US, SI, or other system of measurement) (see 3.1.1).

d. If measuring and indicating devices are required to be graduated in a specific measurement system, stated required system (US, Si, or other system of measurement) (see 3.1.1.1).

e. Additional safety and health requirements, if required (see 3.1.5 and 6.3).

f. Painting, if different (see 3.2.5).

g. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.2.10).

h. Grinding wheel shank sizes 1/4 and 1/8 inch, if required (see 3.3.3).

i. Number, size, and configuration of T-slots, if different (see 3.3.4).

j. Grinding wheel requirements, if different (see 3.3.8).

k. Coolant system, if not required (see 3.3.10).

1. Pneumatic system, if not required (see 3.3.12)*

m. Electrical standards, if different (see 3.3.13).

n. Motor enclosure, if different (see 3.3.13.1).

o. Hour meter, if required (see 3.3.15).

p. Digital readout system, if required (see 3.3.16).

q. Axes to be equipped with digital readout display (see 3.3.16).

r. Standard equipment, if different (see 3.3.18).

s. Size, number, type, characteristics and composition of grinding wheels , if required (see 3.3.18.f).

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t. Optional equipment, if required or if different (see 3.3.19).

u. Type of wheel dresser, if required (see 3.3.19.m)

v. Size and capacities, if different (see 3.4).

w. Repair parts, if required (see 3.7).

x. Lubrication chart, if different (see 3.8.1).

y. Nameplate, if different (see 3.8.2).

z. Quality conformance inspection, if different (see 4.3).

aa. Acceptance test, if different (see 4.4).

bb. Performance test, if different (see 4.6.2).

cc. Alignment and accuracy test requirements, if different (see 4.6.3).

dd. Preservation, packaging, and marking, if different (see 5.1).

6.2.2 Contract data requirements. All data requirements such as parts lists, wiring diagrams, operating manuals, maintenance manuals, and alignment and test results, shall be requested on a DD Form 1423, Contract Data Requirements List, and shall be furnished in accordance with each specified DD Form 1664, Data Item Description, as shown on the DD Form 1423.

6.3 Safety and health requirements. Paragraph 3.1.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment), the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all points-of- operation guarding as required by the manufacturer. The procuring activity should specify the exact configuration of the guard required if as, in most cases, the guard configuration is dependent on the size and configuration of the workplaces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.l(e).

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6.4 Cross-reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

MIL-G-23681B	MIL-G-23681C	
Type I, Size 1124	Type I, Size 1	
N/A	Type II, Style A, Size 1	New
N/A	Type II, Style A, Size 2	New
N/A	Type II, Style A, Size 3	New
N/A	Type II, Style B, Size 1	New
Type II, Style B, Size 2236	Type II, Style B, Size 2	
N/A	Type II, Style B, Size 3	New
Type 11, Style B, Size 2448	Type II, Style B, Size 4	
N/A	Type II, Style B, Size 5	New

6.5 Changes from Previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Army - AV
Navy - SH
Air Force - 84
DLA - GS

User Activities:
Army - None
Navy - MC, OS
Air Force - 99

Preparing Activity:
DLA - IP

Project Number:
3415-0144

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-23681C	Grinding Machines, Jig, Vertical

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

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